

GLOBAL UNIVERSITY ANALYSIS DASHBOARD

1. Project Overview

Objective

The Global University Analysis Dashboard is an interactive Power BI solution designed to analyze worldwide higher education institutions using multiple university ranking datasets. The dashboard provides insights into university performance, country-wise educational excellence, ranking methodologies, student demographics, and yearly trends. It enables users to compare universities, identify leading countries, evaluate ranking systems, and monitor changes in academic performance over time through interactive visualizations.

2. Dataset Used

The dashboard is developed using six interconnected datasets:

- Country
- University
- University Year
- University Ranking Year
- Ranking System
- Ranking Criteria

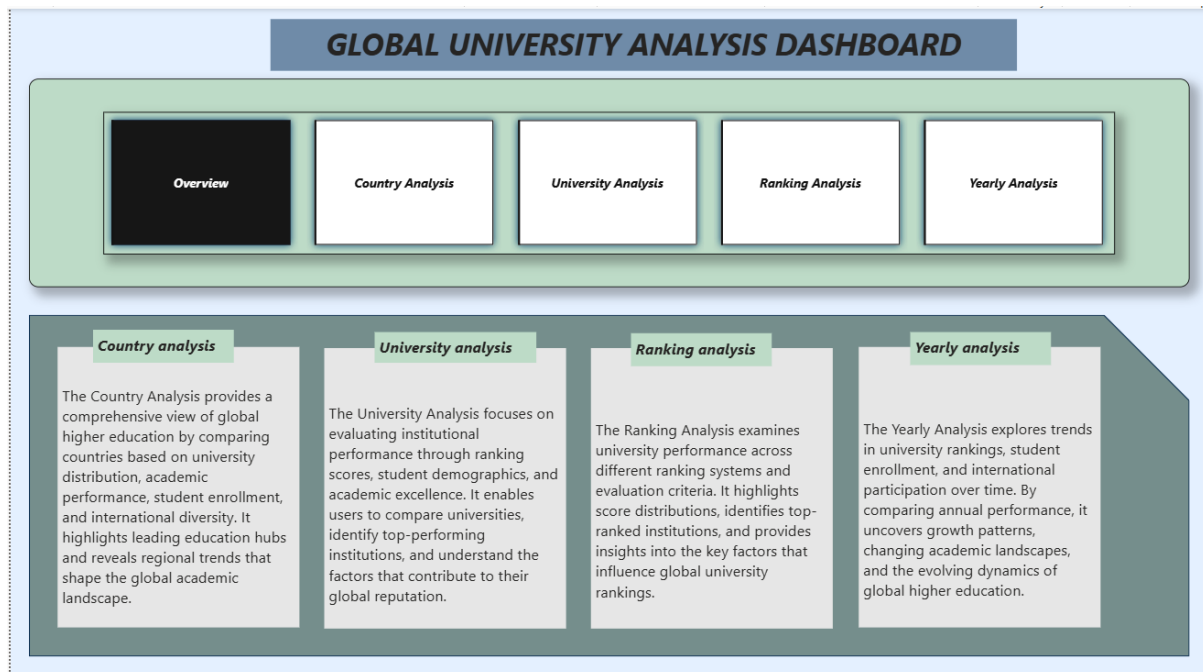
These datasets were modeled using relationships to enable comprehensive analysis across multiple dimensions.

3. Dashboard Structure

The report consists of five interactive pages:

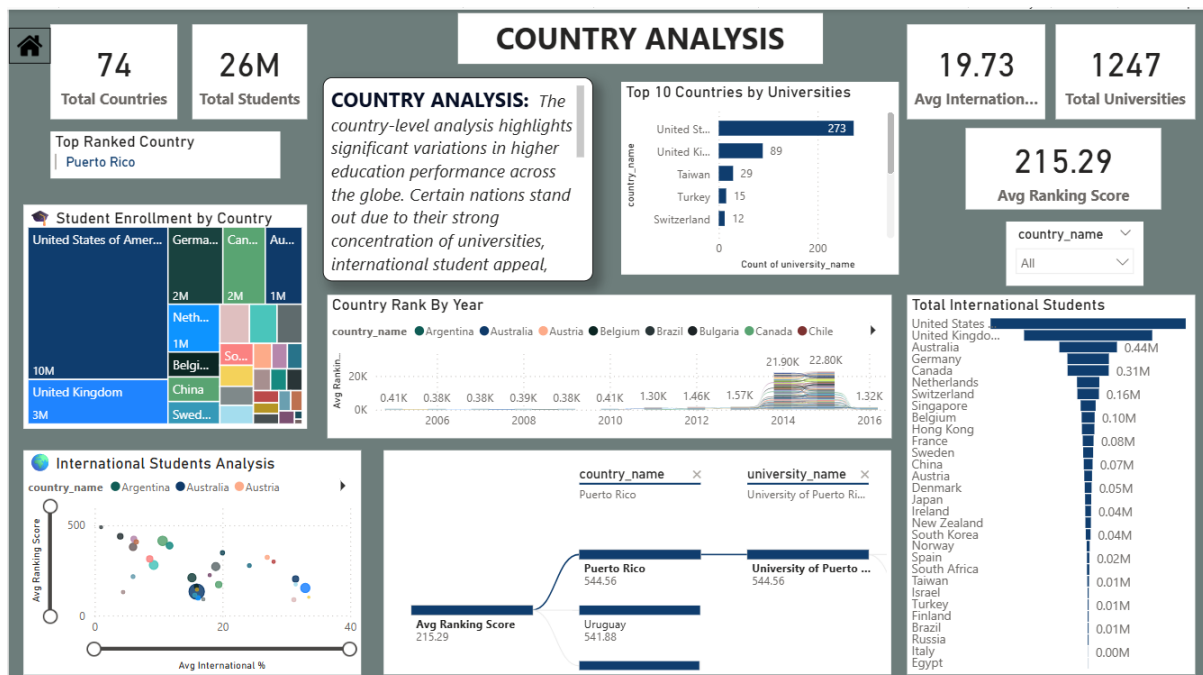
- Overview Dashboard
- Country Performance Analysis
- University Performance Analysis
- Global Ranking Analysis
- Yearly Trends Analysis

Page 1: Overview Dashboard



- **Navigation Panel**
Provides quick access to all dashboard pages, allowing users to seamlessly navigate between different analyses.
- **Country Analysis**
Summarizes country-wise university distribution, academic performance, and student demographics across the globe.
- **University Analysis**
Provides an overview of institutional performance, student diversity, and academic excellence.
- **Ranking Analysis**
Introduces university ranking systems, evaluation criteria, and ranking performance across institutions.
- **Yearly Analysis**
Highlights annual trends in rankings, enrollment, and international student participation over time.

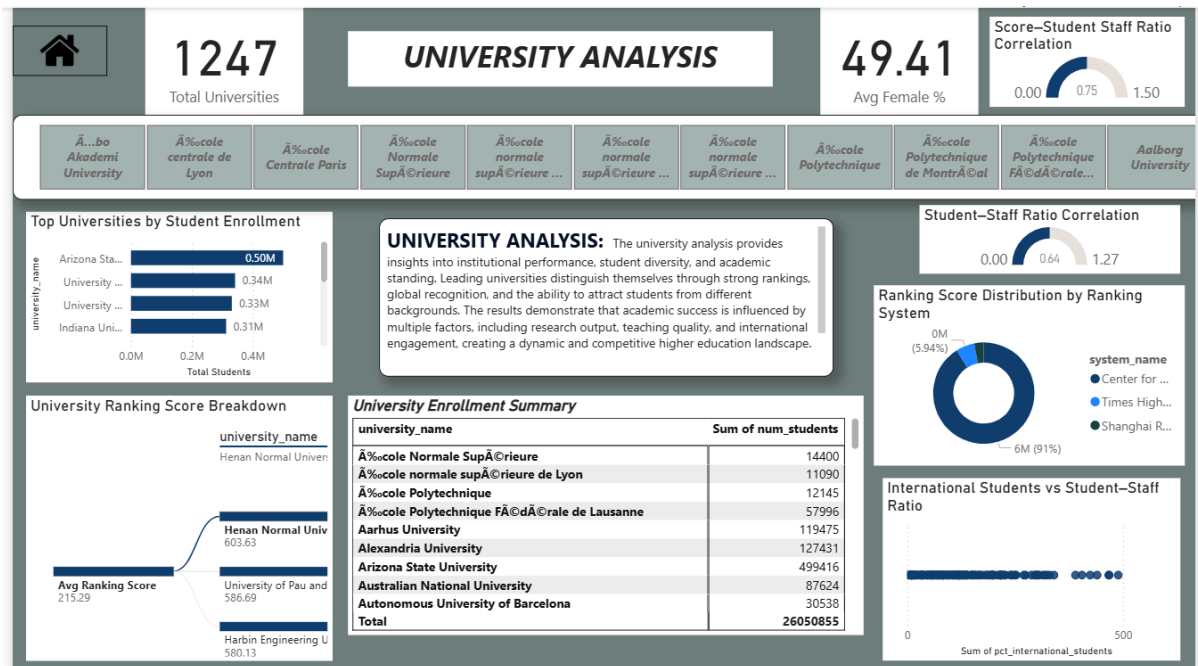
Page 2: Country Performance Analysis



- **Total Countries**
Displays the total number of countries represented in the dataset.
- **Total Students**
Shows the cumulative student population across all countries.
- **Top Ranked Country**
Identifies the country with the highest average university ranking score.
- **Average International Student %**
Displays the average percentage of international students across countries.
- **Total Universities**
Shows the total number of universities included in the analysis.
- **Average Ranking Score**
Represents the average academic ranking score across all countries.
- **Top 10 Countries by Number of Universities**
Compares countries based on the number of universities to identify leading higher education hubs.
- **Top Countries by Student Enrollment**
Illustrates countries with the highest student enrollment across universities.
- **Country Ranking Trends Over Time**
Shows how average ranking scores have changed for different countries across the selected years.
- **International Student Distribution**
Highlights countries attracting the largest number of international students.
- **International Students vs Ranking Score**
Analyzes the relationship between international student percentages and university ranking scores.

- **Ranking Score Breakdown (Decomposition Tree)**
Provides an interactive drill-down of ranking scores by country, university, and year.

Page 3: University Performance Analysis



- **Total Universities**
Displays the total number of universities analyzed.
- **Average Female Student %**
Shows the average female student representation across universities.
- **Student-Staff Ratio Correlation Gauge**
Measures the relationship between student-staff ratio and institutional performance.
- **Ranking Score Correlation Gauge**
Displays the correlation between ranking scores and student-staff ratio.
- **University Selection Slicer**
Allows users to filter all visuals for a specific university.
- **Top Universities by Student Enrollment**
Ranks universities based on their total student population.
- **University Analysis Summary**
Provides an overview of institutional performance and academic excellence.
- **University Enrollment Summary**
Displays total student enrollment for each university in a tabular format.
- **Ranking Score Distribution by Ranking System**
Compares ranking scores across different university ranking systems.

- **International Students vs Student–Staff Ratio**
Examines the relationship between international student percentages and student–staff ratios.
- **University Ranking Breakdown (Decomposition Tree)**
Allows users to explore ranking scores by university and related dimensions.

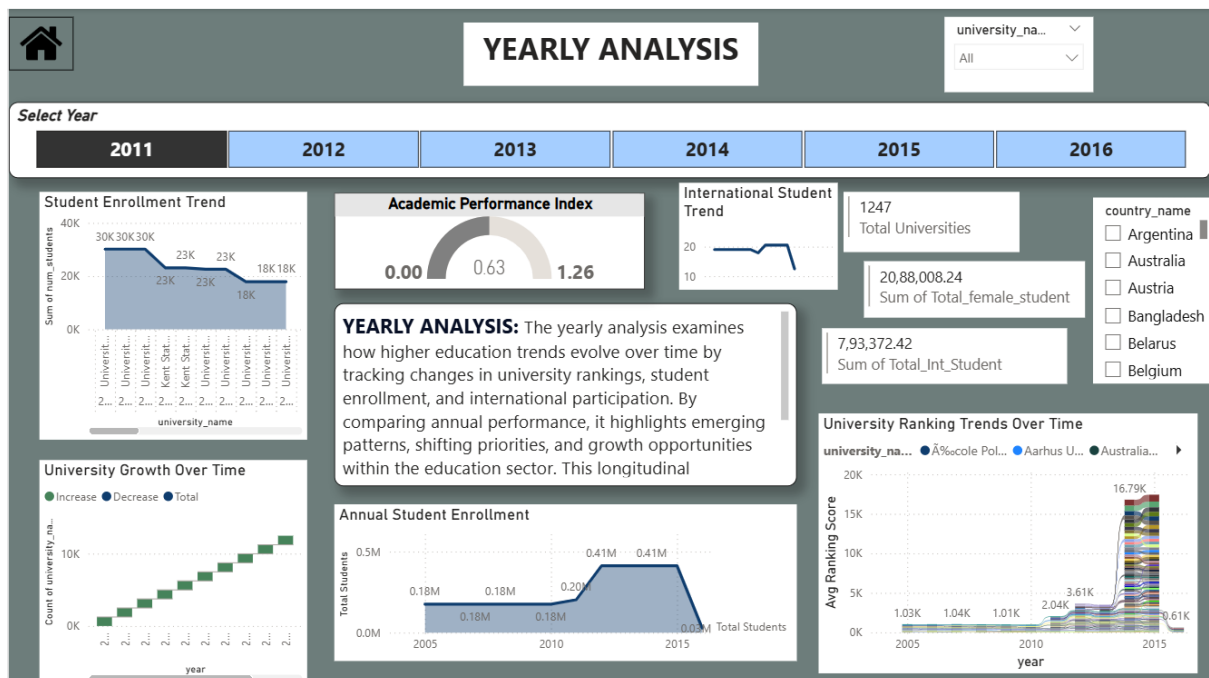
Page 4: Global Ranking Analysis



- **Highest Ranking Score**
Displays the highest ranking score achieved in the dataset.
- **Total Ranking Systems**
Shows the total number of ranking systems included in the analysis.
- **Total Ranking Criteria**
Displays the number of evaluation criteria used in university rankings.
- **Average Ranking Score**
Represents the overall average ranking score across all universities.
- **Top Ranked University**
Identifies the university with the highest average ranking score.
- **Universities by Ranking System**
Shows the distribution of universities across different ranking systems.
- **Academic Excellence Gauge**
Measures the average ranking score against the maximum performance benchmark.
- **Top Universities by Average Ranking Score**
Ranks universities based on their average academic performance.

- **Average Ranking Score by Ranking System**
Compares average ranking scores across various ranking methodologies.
- **Ranking Score Trends by Ranking System**
Tracks how ranking scores change over time for each ranking system.
- **University Ranking Summary**
Provides detailed ranking information for universities in a tabular format.
- **Ranking Score Breakdown (Decomposition Tree)**
Enables interactive exploration of ranking performance across countries, universities, and years.

Page 5: Yearly Trends Analysis



- **Year Selection**
Allows users to analyze university data for a selected year.
- **University Filter**
Filters all dashboard visuals based on the selected university.
- **Student Enrollment Trend**
Shows changes in total student enrollment across different years.
- **Ranking Score vs Student–Staff Ratio**
Measures the correlation between academic performance and student–staff ratio over time.
- **International Student Trend**
Displays yearly changes in the average percentage of international students.
- **Total Universities**
Shows the total number of universities included for the selected period.

- **Total Female Students**
Displays the cumulative number of female students across universities.
- **Total International Students**
Shows the total international student population over the selected years.
- **University Growth Over Time**
Illustrates the increase or decrease in the number of universities across years.
- **Annual Student Enrollment**
Tracks yearly fluctuations in total student enrollment.
- **University Ranking Trends**
Visualizes how university ranking scores evolve across different years.
- **Country Filter**
Allows users to filter yearly analysis based on a selected country.

Business Value

The dashboard provides valuable insights for:

- Students selecting universities
- Educational researchers
- University administrators
- Policy makers
- Government agencies
- Higher education consultants

It enables informed decision-making by presenting a comprehensive view of global higher education performance.

Conclusion

The Global University Analysis Dashboard successfully transforms complex university ranking datasets into an interactive and insightful business intelligence solution. By integrating country-level comparisons, university performance analysis, ranking system evaluation, and longitudinal trend analysis, the dashboard provides a holistic understanding of global higher education.

Its intuitive navigation, interactive filters, dynamic KPIs, and diverse visualizations empower users to explore academic performance from multiple perspectives. The project demonstrates the practical application of Power BI, DAX, data modeling, and data visualization techniques, making it a valuable analytical tool and a strong addition to a professional data analytics portfolio.
